

VAM Split Strategy

Complete Technical Specification

For Independent Audit -- Ravi Mareedu Project

Overview

Parameter	Value
Client	Ravi Mareedu & Sudhir Vyakaranam
Instrument	Leveraged ETF rotation (UPRO = 3x SPY, TQQQ = 3x QQQ)
Data Source	DataBento production daily OHLCV bars
Date Range	Oct 2020 - Dec 2025 (~1,308 trading days)
Initial Capital	\$100,000
Backtest Engine	IFA event-driven backtester with T+1 execution
Prepared By	Insight Fusion Analytics (IFA)
Audit Date	April 2026

Step 1 -- 4-State Machine (UPRO Only)

Step 1 trades a single leveraged ETF (UPRO, 3x S&P 500) using a 4-state machine driven by moving averages, RSI, and VIX. The strategy rotates between full allocation, trimmed positions, defensive mode, and cash based on market conditions.

State Definitions

State	Allocation	Description
BULL_100	99% UPRO	Full bullish -- all signals green
BULL_TRIMMED	50% UPRO	RSI >75 triggered partial sell
DEFENSIVE	50% UPRO	SPY below 50-SMA for 2 consecutive days
CASH	0%	Kill switch active (VIX>30 OR SPY<200-SMA)

Signal Priority (highest to lowest)

Priority	Signal	Rule
1 (HIGHEST)	KILL SWITCH	VIX > 30 OR SPY close < 200-day SMA --> go to CASH immediately
2	DEFENSIVE	SPY close below 50-day SMA for 2 consecutive days --> trim to 50%
3	RSI TRIM	14-day RSI > 75 --> trim to 50%. RSI < 60 --> rebuy to 99%
4 (LOWEST)	RE-ENTRY from CASH	VIX < 30 AND SPY > 200-SMA AND SPY > 50-SMA --> BULL_100

Step 2 -- 6-State Machine (UPRO + TQQQ)

Step 2 extends the strategy to two leveraged ETFs (UPRO + TQQQ) with independent defensive triggers. SPY signals control UPRO allocation; QQQ signals control TQQQ allocation. The kill switch affects both.

State Definitions

State	UPRO Alloc	TQQQ Alloc	Description
BULL_FULL	50%	50%	Both signals bullish
BULL_TRIMMED	25%	25%	RSI trim active
DEF_SPY	0%	50%	SPY defensive, QQQ still bullish
DEF_QQQ	50%	0%	QQQ defensive, SPY still bullish
DEF_BOTH	0%	0%	Both defensive
CASH	0%	0%	Kill switch active

Key Difference from Step 1

SPY and QQQ defensive triggers are **INDEPENDENT**. SPY below its 50-SMA only affects UPRO allocation. QQQ below its 50-SMA only affects TQQQ allocation. The kill switch (VIX>30 OR SPY<200-SMA) affects both instruments simultaneously.

Execution Rules

T+1 Execution

Signal generated at market close (4 PM ET). Trade executed at **NEXT DAY's open price**. Implementation uses a pending_trade queue -- NOT a shift on the price column. This ensures no look-ahead bias.

Split Adjustments

DataBento data is **NOT** split-adjusted in the CSV files. Splits are applied in-memory during backtest execution.

Instrument	Ratio	Date
UPRO	3:1	2018-05-24
UPRO	2:1	2022-01-13
TQQQ	2:1	2021-01-21
TQQQ	2:1	2022-01-13
TQQQ	2:1	2025-11-20

Method: Divide all pre-split OHLC prices by the split ratio.

Dynamic Slippage

Regime	Slippage	Condition
Normal	5 bps (0.05%)	Default market conditions
Stress	20 bps (0.20%)	VIX > 25 OR kill switch fires

Other Parameters

Parameter	Value
Commission	\$1.00 per trade (flat)
Allocation Cap	99% (not 100%) to prevent negative cash balances
Risk-Free Rate	4% annual (subtracted from returns before Sharpe/Sortino)

Claimed Results

Step 1 Results (UPRO Only)

Metric	Value
Final Value	\$274,821
Total Return	174.82%
CAGR	21.44%
Sharpe Ratio	0.676
Sortino Ratio	0.88
Max Drawdown	-38.34% (Mar 10, 2023)
Total Trades	67
Alpha vs SPY	+7.38%

Step 2 Results (UPRO + TQQQ)

Metric	Value
Final Value	\$278,744
Total Return	178.74%
CAGR	21.77%
Sharpe Ratio	0.663
Sortino Ratio	0.861
Max Drawdown	-39.24% (Mar 10, 2023)
Total Trades	180 (90 UPRO + 90 TQQQ)
Alpha vs SPY	+7.71%

Matrix Cross-Check

95.8% directional agreement with normal approach. Matrix is more conservative (lower returns, better drawdown control).

Known Issues & Disclosures

#	Issue	Details
1	Kill switch whipsaw	Mar-Apr 2022 had 4 round trips due to VIX oscillating around 30. No confirmation period on VIX leg by design (per client requirements).
2	2022 drawdown	-38% max drawdown during 2022 bear market. Kill switch limited exposure but did not eliminate losses.
3	T+1 gap risk	Using next-day open assumes execution within first 30 minutes. Covered by slippage assumptions but not exact.
4	DataBento split adjustment	In-memory only. Raw CSV files are NOT adjusted. An auditor re-running the code will get the same results; inspecting raw CSVs will show unadjusted prices.
5	No survivorship bias check	UPRO and TQQQ survived the full backtest period. No delisted instruments were considered.

--- End of Strategy Specification ---

Prepared by Insight Fusion Analytics | April 2026